

GEMINUS ENGINE — MASTER SUBSYSTEM DEPENDENCY MAP & REFERENCE GUIDE

This reference document outlines the exact architecture, data pipelines, and file dependencies across the Geminus game engine. Keep this file in your project root or `docs/` folder to quickly reference which files to inspect when tuning or modifying any system.

1. SUBSYSTEM QUICK-REFERENCE MATRIX

SUBSYSTEM / DOMAIN	LOGIC / ENGINE FILES	DATA & SCHEMA FILES	UI / COMPONENT / VIEW
Map Studio & Content Suite	src/components/Visualizer/MapEditor.jsx src/utils/ReceiverBridge.js src/utils/InteractionFactory.js	public/data/zones/ manifest.json public/data/zones/Z*.json src/config/MasterAssets.js src/config/gdd.js	src/components/ Visualizer/ MapEditor.jsx src/components/ ObjectConfigModal.jsx src/components/ MapMakerBridge.jsx
Combat & Damage Balancing	src/utils/systems.js src/managers/combat/CombatManager.js	src/config/gdd.js src/data/bestiaryData.js src/data/monsterTitles.js	#combat-modal #tab-content-combat public/modules/ battle.html
Monster Forge / Auto-Balancing	src/utils/DevManager.js (forgeZone , forgeEntireBestiary) MapEditor.jsx (generateZoneMobs)	src/data/zonesData.js src/config/gdd.js src/data/bestiaryData.js	Browser Dev Console MapEditor Roster Panel
Player Creation & Profile	src/managers/player/CreationManager.js src/managers/player/ProfileManager.js src/managers/player/StatsManager.js	src/data/racesData.js src/data/alignmentData.js src/data/masteryData.js	Character Creation Modal #tab-content-stats Player Profile Card
Gear, Sockets & Inventory	src/managers/inventory/InventoryManager.js src/managers/inventory/EquipmentManager.js	src/config/gdd.js (items , equipmentSlotConfig) src/data/armoryData.js src/data/jewelryData.js src/data/gemsData.js	#tab-content- inventory #tab-content- equipment
Loot Drops, Shadows & Echoes	src/utils/systems.js (generateLoot , generateShadowLoot , generateEnchantments)	src/data/dropTables.js src/data/ enchantmentData.js src/config/gdd.js	CombatManager.js (Loot log / Floaters)
The Soulforge & Shattering	src/utils/systems.js (rerollEnchantment) src/utils/InteractionBridge.js	src/data/soulforgeData.js src/data/resourceData.js (crafting_resources)	public/modules/ soulforge.html
Economy, Shops & Gilded Vault	src/managers/economy/MerchantManager.js src/managers/economy/VaultManager.js	src/data/vaultData.js src/data/resourceData.js	public/modules/ armory.html public/modules/ arcanum.html public/modules/ gilded_vault.html public/modules/ market.html
Clans & Bastion Upgrades	src/utils/InteractionBridge.js (UPGRADE_BUILDING)	src/data/clanData.js src/data/resourceData.js (war_materials)	public/modules/ clan_hall.html public/modules/ TheClanBastion.html
Soul Debt & Resurrection	src/managers/combat/SanctuaryManager.js src/managers/core/DataManager.js	src/utils/systems.js (Tithe calculation)	public/modules/ sanctuary.html
Social & In-Game Chat	src/managers/social/ChatManager.js	LocalStorage / Firebase State	#footer-chat- container public/modules/ chat.html

SUBSYSTEM / DOMAIN	LOGIC / ENGINE FILES	DATA & SCHEMA FILES	UI / COMPONENT / VIEW
World Map, Zones & Pathfinding	src/managers/world/ZoneManager.js src/managers/world/WorldMapManager.js src/managers/core/GameManager.js	public/data/zones/*.json public/data/zones/manifest.json src/data/zonesData.js	src/managers/world/MapRenderer.js public/player.html
Map Triggers & Bridge Modals	src/config/InteractionRegistry.js src/config/ModuleMap.js src/utils/InteractionBridge.js	src/types/InteractionTypes.js	src/components/InteractionModal.jsx #module-overlay
God Tools / Admin Suite	src/utils/DevManager.js src/stores/StudioStore.js	src/config/gdd.js src/data/gdd_seed.js src/data/*.js	src/components/GodEditor.jsx src/components/Administrator/*Editor.jsx public/devtools.html
In-Game Shell UI & HUD Layout	src/managers/ui/UIManager.js src/managers/ui/LayoutManager.js src/managers/ui/ModalManager.js	src/managers/ui/SettingsManager.js	src/App.jsx src/components/Visualizer/PlayerView.jsx public/player.html

2. CORE ARCHITECTURAL INVARIANTS

A. The Single Source of Truth (SSOT)

- `src/config/gdd.js` flattens all individual data tables into the universal master `items` dictionary.
- Never manually hardcode standalone gear stats inside components or managers without aligning with `gdd.js`.

B. Strict Item Object Schema

Whenever generating, modifying, or transferring items (via `DevManager`, `MerchantManager`, `systems.js`, etc.), the item instance **must** preserve these keys:

```
javascript { id: "BaseRegistryID", uuid: crypto.randomUUID(), instanceId: "GEN_TYPE_TIMESTAMP_RAND", name: "Item Name", tier: 1, qualityMultiplier: 1.0, // 0.75 - 1.5 for Shadows, 0.5 for Echoes sockets: 2, // Strict Schema: Integer capacity socketedGems: [], // Strict Schema: Array of gem instances enchantments: [] // Array of rolled magic affixes }
```

C. Scaling Math & Damage Model

- **Player Damage (Division Curve):** $(90 * \text{Player_WC_or_SC}) / \text{Monster_AC}$
- **Monster Damage (Division Curve):** $(90 * \text{Monster_ATK}) / (\text{Player_AC} * 0.5)$
- **Magic Tier Clamping:** $\text{Math.max}(1, \text{Math.min}(9, \text{Math.ceil}(\text{itemTier} / 2.25)))$

D. MapEditor Internal Engine Protocols

- **Local vs Global State:** `MapEditor.jsx` maintains its own internal Zustand store (`useMapStore`) governing layer grids, tools, spawn points, and active trigger events, while synchronizing with `StudioStore.js` for master data persistence.
- **Auto-Trigger & Interaction Factory:** Placing objects (e.g. `theArmory`, `portalNew`, `sanctuaryReviveNew`) runs through `enrichObjectWithInteraction()`, binding automatic triggers directly to the `triggers` layer with appropriate action payloads (`openModule`, `teleport`).
- **Navigation Baking:** The collision matrix (`bakeCollisionData`) automatically scans both ground terrain (walls, empty void, liquids) and non-interactive obstacles to generate `0` (blocked) or `1` (walkable) grid states for A* pathfinding.

3. THE 3-FILE SHARING RULE

When beginning a new feature or troubleshooting in chat, share only: 1. **The Manager / Logic Controller** (`src/managers/...` or `src/components/Visualizer/MapEditor.jsx`) 2. **The Source Data / Config Table** (`src/data/...` or `src/config/...`) 3. **The UI Component or Module** (`src/components/...` or `public/modules/...`)